Solomonoff Induction

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Prerequisites

- Comfort with discrete probability: conditional distributions, the chain rule, expectations.
- Kullback–Leibler divergence and basic information theory (helpful, developed as needed).

What you'll learn

- Define the Bayesian mixture over a class of environments and prove it is a proper predictor whose posterior updates multiplicatively.
- Prove the cumulative prediction-error bound and specialize it to the Solomonoff prior to recover the Occam bound.
- Show the mixture makes boundedly many mistakes and is Pareto-optimal under KL and squared loss.
- Bound the misspecified case when the truth lies outside the model class.

Much of this material is drawn from [HQC24, Chapter 3] and the earlier [Hut05]; both books provide fuller explanations, additional context, and proofs that we gloss over here.

Difficulty ratings. Each problem is tagged with a rating in square brackets following Knuth's exercise rating scheme [Knu73]: roughly, **[00]** trivial, **[10]** 15–minute pencil-and-paper, **[20]** 1–2 hours, **[30]** several hours to a day, **[40]** a significant research result. Intermediate values are possible.

Overview

Solomonoff induction is the *prediction* half of **Universal AI**: what would an *optimally intelligent* agent do if it only had to predict, given unlimited compute and the weakest possible assumptions? The answer is a single **Bayesian mixture** ξ over a countable class $\mathcal{M}$ of candidate environments, weighted by a prior w_ν. It (eventually) predicts as well as the true environment μ, and under the **universal** choice — $\mathcal{M}$ = all computable environments, prior $w_\nu = 2^{-K(\nu)}$ with K the Kolmogorov complexity — the assumption "$\mu \in \mathcal{M}$" becomes "the universe is computable," and Occam's razor drops out of the mathematics.

This module is a worksheet: you build the theory yourself, one problem at a time. The sequel module, AIXI, lifts the same mixture to sequential decision-making (learning to *act*).

Goal and Roadmap

A recipe for prediction: Solomonoff induction is an attempt to mathematically formalize the *problem of induction* from philosophy: How to make predictions about the future based on past observations?

We formalize this as sequence prediction: There is some true environment μ that generates a sequence $x_1, x_2, \ldots$ of (binary) symbols. Each next symbol $x_t \sim \mu(\cdot \mid x_{<t})$ is sampled from μ conditioned on the past $x_{<t}$. A predictor P takes the history $x_{<t}$ and gives a distribution $P(\cdot \mid x_{<t})$ over the next symbol x_t.

We measure the quality of a predictor P by the μ-expected squared prediction error, summed over every timestep:

$$S_\infty^\mu := \sum_{t=1}^{\infty} \sum_{x_{<t} \in \mathbb{B}^{t-1}} \mu(x_{<t}) \sum_{x_t \in \mathbb{B}} \big(P(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t})\big)^2. \tag{1}$$

How to construct a predictor P such that S_∞^μ is small (or at least finite)? Bayesian inference to the rescue: we choose as our predictor a *Bayesian mixture* ξ over a countable class $\mathcal{M} = \{\nu_1, \nu_2, \ldots\}$ of candidate environments, weighted by prior beliefs w_ν (formalized in Definition 1.2). The main results we build up to are:

- **Cumulative bound** (Section 4): Assuming $\mu \in \mathcal{M}$, $S_\infty^\mu \leq -\ln w_\mu$. The higher the prior on μ, the lower the prediction error.
- **Explicit bound** (Exercise 4.2): specialize to the Solomonoff prior $w_\nu = 2^{-K(\nu)}$ to get $S_\infty^\mu \leq K(\mu) \ln 2$, where K is the *Kolmogorov complexity*.

- **Pareto optimality** (Exercises 6.2 and 8.2): no other predictor weakly dominates ξ on every $\nu \in \mathcal{M}$, for either KL or squared loss.
- **Misspecified version** (Section 7): If $\mu \notin \mathcal{M}$, the cumulative bound becomes $-\ln w_{\hat{\mu}} + D_n(\mu \parallel \hat{\mu})$: the constant complexity term plus an approximation term $D_n(\mu \parallel \hat{\mu}) = \sum_{t=1}^{n} d_t(\mu \parallel \hat{\mu})$ that in general grows linearly in n, so S_∞^μ diverges. Here $\hat{\mu} \in \mathcal{M}$ is the "closest" environment to μ.

Notation

For more background, see the corresponding post on Solomonoff induction.

Note

Symbols used throughout.

- $\mathbb{B} = \{0, 1\}$: the binary alphabet; $\mathbb{B}^*$ all finite binary strings; $\mathbb{B}^n$ length-n strings.
- ϵ: the empty string.
- xy: string concatenation. if $x := x_{1:n}$ and $y := y_{1:m}$ then $xy = x_1 \dots x_n y_1 \dots y_m$.
- $x_{i:j} := x_i x_{i+1} \dots x_j$.
- $x_{<t} := x_1 x_2 \cdots x_{t-1}$.
- ν, ρ: generic environments / predictors. μ: the true (unknown) environment generating the data. ξ: a Bayesian mixture of environments.
- $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$: a countable class of candidate environments.
- $w_\nu > 0$ with $\sum_{\nu \in \mathcal{M}} w_\nu = 1$: the prior over $\mathcal{M}$. $w(\nu \mid x_{<t})$: the posterior after observing $x_{<t}$.
- $\Delta\mathcal{X}$: the set of probability distributions over a finite set $\mathcal{X}$.

1. The mixture is a predictor

Definition 1.1 (Environment)**.** An **environment** ν assigns to each history $x_{<t} \in \mathbb{B}^*$ a predictive distribution $\nu(\cdot \mid x_{<t}) \in \Delta\mathbb{B}$ over the next symbol. The conditional joint probability of a string $x \in \mathbb{B}^*$, conditioned on $y \in \mathbb{B}^*$, is defined

as

$$\nu(x_{1:n}|y) \;:=\; \prod_{t=1}^{n} \nu(x_t \mid y x_{<t}), \qquad \nu(\epsilon) := 1.$$

The unconditional joint is defined as $\nu(x_{1:n}) := \nu(x_{1:n} \mid \epsilon)$. The true (unknown) environment generating the data is denoted μ.

Remark

One can show that $\nu(x_{1:t}) = \nu(x_{<t}) \cdot \nu(x_t \mid x_{<t})$ and $\nu(x) = \nu(x0) + \nu(x1)$ for all $x \in \mathbb{B}^*$.

Definition 1.2 (Bayesian mixture ξ)**.** Let $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ be a countable class of environments with prior weights $w_\nu > 0$ satisfying $\sum_{\nu \in \mathcal{M}} w_\nu = 1$. The **Bayesian mixture** is defined as a prior-weighted mixture over all environments ν in $\mathcal{M}$:

$$\xi(x_{1:t}) \;:=\; \sum_{\nu \in \mathcal{M}} w_\nu \, \nu(x_{1:t}). \tag{2}$$

Its one-step predictive distribution is

$$\xi(x_t \mid x_{<t}) \;=\; \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \, \nu(x_t \mid x_{<t}),$$

with posterior weights

$$w(\nu \mid x_{<t}) \;:=\; w_\nu \, \frac{\nu(x_{<t})}{\xi(x_{<t})} \qquad w(\nu \mid \epsilon) := w_\nu.$$

Here $w(\nu \mid x_{<t})$ is the *posterior* belief in ν after observing $x_{<t}$.

The mixture ξ is defined as a sum of *joint* probabilities. To use it as a predictor we need its one-step conditional $\xi(x_t \mid x_{<t})$, and we need to know it is a genuine probability distribution (so that, later, Pinsker's inequality applies to it).

Exercise 1.1 (Generalized chain rule). [05]
The two-term chain rule baked into Definition 1.1 extends to arbitrary contiguous blocks. Show that for any $1 \le p \le q \le r$ and any history $x_{<p}$ with $\nu(x_{<p}) > 0$,

$$\nu(x_{p:r} \mid x_{<p}) \;=\; \nu(x_{p:q} \mid x_{<p}) \cdot \nu(x_{q+1:r} \mid x_{<p}\, x_{p:q}).$$

Exercise 1.2 (Mixture is a predictor). [10]

Starting from $\xi(x_{1:t}) = \sum_{\nu \in \mathcal{M}} w_\nu \, \nu(x_{1:t})$, show that

$$\xi(x_t \mid x_{<t}) \;=\; \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \, \nu(x_t \mid x_{<t}), \qquad w(\nu \mid x_{<t}) := w_\nu \, \frac{\nu(x_{<t})}{\xi(x_{<t})},$$

and conclude that $\sum_{x_t \in \mathbb{B}} \xi(x_t \mid x_{<t}) = 1$, i.e. $\xi(\cdot \mid x_{<t})$ is a probability distribution over $\mathbb{B}$.

Hint: Write $\xi(x_t \mid x_{<t}) = \xi(x_{1:t})/\xi(x_{<t})$, expand the numerator, and use the chain rule $\nu(x_{1:t}) = \nu(x_{<t}) \, \nu(x_t \mid x_{<t})$ (Definition 1.1). For the last part, note that the posterior weights sum to 1.

Exercise 1.3 (Posterior update). [10]
Show that the posterior weight $w(\cdot \mid \cdot)$ updates *multiplicatively* as new symbols arrive. For every $x_{1:t} \in \mathbb{B}^*$ with $\xi(x_{1:t}) > 0$,

$$w(\nu \mid x_{1:t}) \;=\; w(\nu \mid x_{<t}) \cdot \frac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}.$$

That is: the new posterior equals the old posterior, scaled by the **likelihood ratio** of how well ν predicted the just-observed symbol relative to the mixture's own prediction.

Exercise 1.4 (Multi-step posterior linearity). [10]
Exercise 1.2 showed that ξ's one-step prediction $\xi(x_t \mid x_{<t})$ is the posterior-weighted average of the per-model one-step predictions. The same identity extends to predictions over a *whole future segment* $x_{t:m}$: for any $t \le m$,

$$\xi(x_{t:m} \mid x_{<t}) \;=\; \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \, \nu(x_{t:m} \mid x_{<t}).$$

That is: the *same* posterior $w(\nu \mid x_{<t})$ governs ξ's predictions for arbitrarily many steps into the future, not just the next one. Show this.

Hint: Write $\xi(x_{t:m} \mid x_{<t}) = \xi(x_{1:m})/\xi(x_{<t})$, expand the numerator using the mixture definition, and apply the generalized chain rule (Exercise 1.1 with $p = 1$, $q = t-1$, $r = m$) to factor each $\nu(x_{1:m}) = \nu(x_{<t}) \, \nu(x_{t:m} \mid x_{<t})$.

2. KL divergence

The KL divergence between two joint distributions over a length-n history splits, by an inductive application of the chain rule for probabilities, into a sum of per-step conditional KLs. This telescoping identity lets us trade a single joint KL over an entire history for a sum of per-step KLs (and vice versa): exactly the bridge we will need in Section 4.

We introduce two pieces of notation that will be used throughout the rest of the worksheet:

Definition 2.1 (Cumulative and per-step KL)**.** For environments / predictors ν, ρ, horizon $n \geq 1$, and $1 \leq t \leq n$, the **cumulative joint KL** and the **expected per-step KL at step** t are

$$D_n(\nu \parallel \rho) \;:=\; \sum_{x_{1:n} \in \mathbb{B}^n} \nu(x_{1:n}) \ln \frac{\nu(x_{1:n})}{\rho(x_{1:n})},$$

$$d_t(\nu \parallel \rho) \;:=\; \sum_{x_{<t} \in \mathbb{B}^{t-1}} \nu(x_{<t}) \sum_{x_t \in \mathbb{B}} \nu(x_t \mid x_{<t}) \ln \frac{\nu(x_t \mid x_{<t})}{\rho(x_t \mid x_{<t})},$$

with the conventions $0 \ln \frac{0}{q} := 0$ for $q \geq 0$ and $p \ln \frac{p}{0} := \infty$ for $p > 0$. We write $D_\infty(\nu \parallel \rho) := \lim_{t \to \infty} D_t(\nu \parallel \rho)$ and $D_\infty^\mu := D_\infty(\mu \parallel \xi)$.

Theorem 2.2 (KL divergence is non-negative)**.** *For any environments ν, ρ and any horizon $n \geq 1$ and step $1 \leq t \leq n$:*

(a) $D_n(\nu \parallel \rho) \geq 0$, *with equality iff* $\nu(x_{1:n}) = \rho(x_{1:n})$ *for every* $x_{1:n} \in \mathbb{B}^n$ *with* $\nu(x_{1:n}) > 0$.

(b) $d_t(\nu \parallel \rho) \geq 0$, *with equality iff* $\nu(x_t \mid x_{<t}) = \rho(x_t \mid x_{<t})$ *for every* $x_{<t}$ *with* $\nu(x_{<t}) > 0$ *and every* $x_t \in \mathbb{B}$.

Proof in Appendix B.

Exercise 2.1 (Telescoping KL). [10]
Show by induction on n that for all environments ν, ρ,

$$D_n(\nu \parallel \rho) \;=\; \sum_{t=1}^{n} d_t(\nu \parallel \rho),$$

where on the right-hand side, the $t = 1$ summand uses the conventions $\nu(x_1 \mid x_{<1}) := \nu(x_1)$, $\rho(x_1 \mid x_{<1}) := \rho(x_1)$.

Hint: For the induction step, factor $\nu(x_{1:n}) = \nu(x_{<n})\,\nu(x_n \mid x_{<n})$ (and likewise for ρ) inside the log; the log splits additively into two pieces matching $D_{n-1}(\nu \,\|\, \rho)$ and $d_n(\nu \,\|\, \rho)$.

3. Mixture dominance

The mixture ξ never assigns much less probability than any single environment weighted by its prior. This single inequality is the engine behind the cumulative bound of Section 4 below: dividing through gives $\mu(x)/\xi(x) \le 1/w_\mu$, which is then fed into Pinsker and the chain rule in the proof of the main cumulative bound.

Exercise 3.1 (Mixture dominance). [05]
Show that for every $\nu \in \mathcal{M}$ and every $x \in \mathbb{B}^*$,

$$\xi(x) \;\ge\; w_\nu \cdot \nu(x).$$

Remark (Solomonoff specialization)

Under the Solomonoff prior, $w_\nu = 2^{-K(\nu)}$, so the bound becomes

$$\xi_U(x) \;\ge\; 2^{-K(\nu)} \cdot \nu(x) \qquad \text{for every computable } \nu.$$

This is what makes ξ_U *universal*: a single predictor dominates the entire computable model class up to a factor that depends only on the model's description length.

4. Cumulative prediction error bound (main result)

The next ingredient relates squared prediction error to KL divergence [CT06, Lemma 11.6.1]. We require the following inequality:

Theorem 4.1 (Pinsker's inequality)**.** *Let P and Q be probability distributions over $\mathbb{B} = \{0, 1\}$, i.e., $p_0, p_1, q_0, q_1 \ge 0$ with $p_0 + p_1 = 1$ and $q_0 + q_1 = 1$. Write $p_x = P(x)$ and $q_x = Q(x)$. Then*

$$\sum_{x \in \mathbb{B}} \big(q_x - p_x\big)^2 \;\le\; \sum_{x \in \mathbb{B}} p_x \ln \frac{p_x}{q_x}. \tag{$\star$}$$

The proof of this inequality is mostly tedious algebra but is included in Appendix A.

Definition 4.2 (Cumulative expected and per-step squared prediction error)**.** For environments / predictors ν, ρ and horizon $n \geq 1$, the **cumulative expected squared error** and the **per-step squared error at step** t are

$$S_n(\nu \parallel \rho) := \sum_{t=1}^{n} s_t(\nu \parallel \rho)$$
$$s_t(\nu \parallel \rho) := \sum_{x_{<t} \in \mathbb{B}^{t-1}} \nu(x_{<t}) \sum_{x_t \in \mathbb{B}} \big(\nu(x_t \mid x_{<t}) - \rho(x_t \mid x_{<t})\big)^2$$

Write $S_\infty(\nu \parallel \rho) := \lim_{n\to\infty} S_n(\nu \parallel \rho)$ and $S_\infty^\mu := S_\infty(\mu \parallel \xi)$.

Exercise 4.1 (Cumulative prediction bound). [15]
Show that

$$S_\infty^\mu \leq -\ln w_\mu.$$

Hint: You need Theorem 4.1, Exercise 2.1 and Exercise 3.1.

Exercise 4.2 (Explicit complexity bound). [05]
Specialize Section 4 to the **Solomonoff prior** $w_\nu = 2^{-K(\nu)}$ (where $K(\nu)$ is the length of the shortest program computing ν) to show

$$S_\infty^\mu \leq K(\mu) \ln 2.$$

Exercise 4.3 (Per-step error). [05]
Hence prove that the per-step expected prediction error

$$s_t := \sum_{x_{<t} \in \mathbb{B}^{t-1}} \mu(x_{<t}) \sum_{x_t \in \mathbb{B}} \big(\xi(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t})\big)^2$$

converges to zero as $t \to \infty$.

5. Bounded number of prediction mistakes

So far we have bounded *squared error*. For a deterministic environment (i.e. a single infinite binary sequence $x_1^*, x_2^*, \ldots$ in $\mathcal{M}$), one can ask the sharper 0/1-loss question: how often does the mixture predict the *wrong bit*? The cumulative bound is strong enough to give an answer.

Let $\mu \in \mathcal{M}$ be a deterministic environment, so $\mu(x_t \mid x_{<t}) \in \{0, 1\}$ on every history;

write x_t^* for the unique bit on which $\mu(\cdot \mid x_{<t}^*)$ puts all its mass. Define the **threshold predictor** associated with ξ: at each step, predict the more-likely bit,

$$\hat{x}_t \;:=\; \arg\max_{x \in \mathbb{B}} \xi(x \mid x_{<t}) \qquad \text{(break ties arbitrarily)}.$$

We say the mixture makes a **mistake** at time t if $\hat{x}_t \neq x_t^*$.

Exercise 5.1 (Bounded prediction mistakes). [15]
Show that the number of mistakes made by the threshold predictor on the μ-trajectory is at most

$$\#\{\, t \geq 1 \;:\; \hat{x}_t \neq x_t^* \,\} \;\leq\; -2 \ln w_\mu.$$

Hint: At a mistake step, $\xi(x_t^* \mid x_{<t}) \leq \frac{1}{2}$ (why?). The per-step squared error along the deterministic μ-trajectory simplifies dramatically; bound it from below, then sum.

Remark (Solomonoff specialization)

Under the Solomonoff prior $w_\mu = 2^{-K(\mu)}$, so $-\ln w_\mu = K(\mu) \ln 2$ and the bound becomes

$$\#\{\, t \geq 1 \;:\; \hat{x}_t \neq x_t^* \,\} \;\leq\; 2K(\mu) \ln 2.$$

With $\ln 2 \approx 0.693$ the constant is just under 1.4, so if the true environment μ is computable, and can be described by a program at most k bits long, then the predictor ξ will make at worst $\approx 1.4k$ mistakes over predicting the entire sequence $x_{1:\infty}^*$. Note that the bound is purely existence-style: it does not say *when* the mistakes happen. They could all occur at the start, or be arbitrarily far into the future.

6. Pareto optimality under KL loss

In Section 6 and Section 8 we establish that the Bayesian mixture ξ is *uniquely Pareto-optimal* under both KL and squared prediction loss: no other predictor can do at least as well as ξ on every environment $\nu \in \mathcal{M}$ without coinciding with ξ entirely.

Definition 6.1 (Pareto domination)**.** For some measure of **loss** $\mathcal{F}(\nu, \rho)$ that takes an environment ν and a predictor ρ, a predictor ρ **weakly Pareto-dominates** ξ with respect to loss $\mathcal{F}$ and class $\mathcal{M}$ if

$$\mathcal{F}(\nu, \rho) \;\leq\; \mathcal{F}(\nu, \xi) \qquad \text{for every } \nu \in \mathcal{M}.$$

ξ is **Pareto-optimal** (w.r.t. $\mathcal{F}$) if the only predictor that weakly dominates it

is ξ itself.
We specialize $\mathcal{F}$ to $D_n(\nu \,\|\, \rho)$ in Section 6 and to $S_n(\nu \,\|\, \rho)$ in Section 8.

Exercise 6.1 (KL Pythagorean identity). [10]
For *any* predictor ρ with $\rho(x_{1:n}) > 0$ on every $x_{1:n} \in \mathbb{B}^n$, show that

$$D_n(\xi \,\|\, \rho) \;+\; \sum_{\nu \in \mathcal{M}} w_\nu \, D_n(\nu \,\|\, \xi) \;=\; \sum_{\nu \in \mathcal{M}} w_\nu \, D_n(\nu \,\|\, \rho).$$

Hint: Take the difference between the two summation terms.

Exercise 6.2 (KL Pareto-optimality). [05]
Show that ξ is Pareto-optimal under $D_n(\nu \,\|\, \rho)$ in the sense of Definition 6.1: if ρ is any predictor with

$$D_n(\nu \,\|\, \rho) \;\le\; D_n(\nu \,\|\, \xi) \qquad \text{for every } \nu \in \mathcal{M},$$

then $\rho = \xi$.

Hint: Multiply the assumed inequality by $w_\nu \ge 0$ and sum over $\nu \in \mathcal{M}$; then apply the KL Pythagorean identity to recognize the extra non-negative term, which must vanish.

7. Misspecified models: when $\mu \notin \mathcal{M}$

The cumulative bound Section 4 assumed the true environment μ lives in the model class $\mathcal{M}$. What happens when it does not? We now show the bound *degrades gracefully*: it splits cleanly into a complexity term ("cost of not knowing which model is best"), exactly as before, plus a linear-in-time approximation term ("cost of no model being right"). See [Hut05, §3.2.8] for the original treatment.

Throughout this section, we no longer assume $\mu \in \mathcal{M}$. We will state the bound in terms of an arbitrary $\hat{\mu} \in \mathcal{M}$, so taking the infimum over $\hat{\mu}$ on the right-hand side gives the tightest version by using the "closest" approximation of $\hat{\mu} \in \mathcal{M}$ to μ.

Exercise 7.1 (Misspecified KL bound). [10]
Show that for all $\hat{\mu} \in \mathcal{M}$,

$$D_n(\mu \,\|\, \xi) \;\le\; -\ln w_{\hat{\mu}} \;+\; D_n(\mu \,\|\, \hat{\mu}).$$

Hint: Use Exercise 3.1.

Remark (Reading the two terms)

The right-hand side splits into two qualitatively different terms:

- $-\ln w_{\hat{\mu}}$ is *constant in* n: under the Solomonoff prior, this is $K(\hat{\mu})\ln 2$. The familiar "complexity" cost of search.
- The approximation term $D_n(\mu \parallel \hat{\mu}) = \sum_{t=1}^{n} d_t(\mu \parallel \hat{\mu})$ (by Exercise 2.1) is a sum of n per-step KLs; in general it grows with n. It vanishes identically iff $\hat{\mu}$ matches μ on every μ-reachable history, in particular when $\mu \in \mathcal{M}$ and we pick $\hat{\mu} = \mu$, recovering Section 4.

Combining with Pinsker (Theorem 4.1) gives the corresponding bound on S_∞^μ, which is infinite in general but inherits the rate of $\hat{\mu}$.

Details on how to define the best choice of $\hat{\mu}$ are in Appendix D.

8. Pareto optimality under squared loss

The squared specialization $\mathcal{F}(\nu, \rho) = S_n(\nu \parallel \rho)$ (Definition 4.2) lives one level down from KL: at the conditional distributions $\nu(\cdot \mid x_{<t})$. The natural Pythagorean decomposition at a fixed history uses *posterior* weights $w(\nu \mid x_{<t})$: those are the weights that make $\xi(x_t \mid x_{<t})$ the mean of the $\nu(x_t \mid x_{<t})$'s (via the posterior-predictive form, Definition 1.2). The Pareto-optimality aggregation, however, uses prior weights w_ν. Bridging the two takes one extra Bayes-rule step.

Exercise 8.1 (Squared Pythagorean identity). [10]
Fix any $t \in \{1, \dots, n\}$ and any history $x_{<t} \in \mathbb{B}^{t-1}$: *the history is arbitrary, not sampled from any environment.* For any $x_t \in \mathbb{B}$ and any predictor ρ, show

$$\big(\xi(x_t \mid x_{<t}) - \rho(x_t \mid x_{<t})\big)^2 + \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \big(\nu(x_t \mid x_{<t}) - \xi(x_t \mid x_{<t})\big)^2$$
$$= \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \big(\nu(x_t \mid x_{<t}) - \rho(x_t \mid x_{<t})\big)^2.$$

Hint: Add and subtract $\xi(x_t \mid x_{<t})$ inside the squared term, then expand. Use $\sum_\nu w(\nu \mid x_{<t}) = 1$ and the posterior-predictive form of ξ (Definition 1.2), $\sum_\nu w(\nu \mid x_{<t})\, \nu(x_t \mid x_{<t}) = \xi(x_t \mid x_{<t})$, to kill the cross-term. The weighting must be *posterior* weights $w(\nu \mid x_{<t})$ (not prior weights w_ν), because $\xi(x_t \mid x_{<t})$ is the posterior-weighted mean of the $\nu(x_t \mid x_{<t})$'s, not the prior one.

Exercise 8.2 (Squared Pareto-optimality). [05]
Show that ξ is Pareto-optimal under $S_n(\nu \parallel \rho)$ in the sense of Definition 6.1: if ρ is any predictor with

$$S_n(\nu \parallel \rho) \;\leq\; S_n(\nu \parallel \xi) \qquad \text{for every } \nu \in \mathcal{M},$$

then $\rho(x_t \mid x_{<t}) = \xi(x_t \mid x_{<t})$ at every history $x_{<t} \in \mathbb{B}^{t-1}$ *with* $\xi(x_{<t}) > 0$ (equivalently, ξ-almost surely) and every $x_t \in \mathbb{B}$. Histories that the mixture never reaches ($\xi(x_{<t}) = 0$, i.e., reached by no $\nu \in \mathcal{M}$ either) are invisible to the loss and are not pinned down.

Hint: Multiply the assumed inequality by $w_\nu \geq 0$ and sum over ν; then for each fixed history $x_{<t}$, multiply the per-history Pythagorean identity from above by $\xi(x_{<t})$ and use the Bayes identity $w(\nu \mid x_{<t})\,\xi(x_{<t}) = w_\nu\,\nu(x_{<t})$ to bridge from posterior weights (inside the identity) to prior weights (in the aggregated Pareto inequality).

A. Proof of Pinsker's inequality

We first prove the following lemma:

Lemma A.1 (Pinsker's binary inequality)**.** *For $0 \le p \le 1$ and $0 < q < 1$ we have*

$$2(p-q)^2 \;\le\; p \ln \frac{p}{q} + (1-p) \ln \frac{1-p}{1-q}.$$

Define $f(x) := p \ln x + (1-p) \ln(1-x)$, so that

$$p \ln \frac{p}{q} + (1-p) \ln \frac{1-p}{1-q} = f(p) - f(q) = \int_q^p f'(x)\, dx = \int_q^p \frac{p-x}{x(1-x)}\, dx,$$

using $f'(x) = \frac{p}{x} - \frac{1-p}{1-x} = \frac{p-x}{x(1-x)}$. Recall also that $x(1-x) \le \frac{1}{4}$ on $[0,1]$, so $\frac{1}{x(1-x)} \ge 4$.

Case $q \le p$. For every $x \in [q,p]$ we have $x \le p$, hence $p - x \ge 0$.

$$\frac{p-x}{x(1-x)} \;\ge\; 4(p-x) \qquad \text{for } x \in [q,p].$$

Integrating over $[q,p]$,

$$\int_q^p \frac{p-x}{x(1-x)}\, dx \;\ge\; 4 \int_q^p (p-x)\, dx \;=\; 2(p-q)^2.$$

Case $q \ge p$. For every $x \in [p,q]$ we have $x \ge p$, hence $x - p \ge 0$.

$$\frac{x-p}{x(1-x)} \;\ge\; 4(x-p) \qquad \text{for } x \in [p,q].$$

Integrating over $[p,q]$,

$$\begin{aligned} \int_q^p \frac{p-x}{x(1-x)}\, dx &= \int_p^q \frac{x-p}{x(1-x)}\, dx \\ &\ge 4 \int_p^q (x-p)\, dx = 4 \int_q^p (p-x)\, dx = 2(p-q)^2 \end{aligned}$$

Proof of Theorem 4.1. We now reduce Theorem 4.1 to the binary inequality Lemma A.1. Recall $p_0 + p_1 = 1$ and $q_0 + q_1 = 1$. We handle the boundary case $q_x = 0$ first, then reduce the interior case directly to Lemma A.1.

Case 1: $q_x = 0$ for some $x \in \mathbb{B}$. If $p_x > 0$, then the right-hand side of $(\star)$ contains $p_x \ln \frac{p_x}{0} = +\infty$, so $(\star)$ holds trivially. If instead $p_x = 0$, that atom contributes 0 to both sides (using the convention $0 \ln \frac{0}{0} = 0$). The other atom x' then satisfies $p_{x'} = 1$ and $q_{x'} = 1 - q_x = 1$, so both sides vanish and $(\star)$ holds.

Case 2: $q_0, q_1 > 0$**.** Since $q_1 = 1 - q_0$ and $p_1 = 1 - p_0$, the two sides of $(\star)$ become

$$\sum_{x \in \mathbb{B}} (q_x - p_x)^2 = (q_0 - p_0)^2 + \big((1 - q_0) - (1 - p_0)\big)^2 = 2(p_0 - q_0)^2,$$

$$\sum_{x \in \mathbb{B}} p_x \ln \tfrac{p_x}{q_x} = p_0 \ln \tfrac{p_0}{q_0} + (1 - p_0) \ln \tfrac{1-p_0}{1-q_0}.$$

So $(\star)$ is exactly $2(p_0 - q_0)^2 \le p_0 \ln \frac{p_0}{q_0} + (1 - p_0) \ln \frac{1-p_0}{1-q_0}$, which is Lemma A.1 with $p = p_0$ and $q = q_0$ (here $0 < q_0 < 1$, since $q_0, q_1 > 0$). □

B. Proof of KL non-negativity

We prove Theorem 2.2. The core is the elementary inequality

$$\ln x \;\le\; x - 1 \qquad \text{for all } x > 0, \quad \text{equality iff } x = 1.$$

(Let $f(x) := (x - 1) - \ln x$; then $f'(x) = 1 - 1/x$, vanishing only at $x = 1$, with $f''(x) = 1/x^2 > 0$, so f is strictly convex and attains its unique minimum $f(1) = 0$ at $x = 1$.)

Lemma B.1 (Gibbs' inequality)**.** *Let P, Q be probability distributions over a finite set $\mathcal{X}$, with the conventions $0 \ln \frac{0}{q} := 0$ and $p \ln \frac{p}{0} := \infty$. Then*

$$\sum_{x \in \mathcal{X}} P(x) \ln \frac{P(x)}{Q(x)} \;\ge\; 0,$$

with equality iff $P(x) = Q(x)$ for every x with $P(x) > 0$.

Proof. If $Q(x) = 0$ for some x with $P(x) > 0$, the LHS is $+\infty$ and the inequality is strict. Otherwise restrict the sum to $\{x : P(x) > 0\}$ (terms with $P(x) = 0$ contribute 0 by convention), where every ratio $Q(x)/P(x)$ is in $(0, \infty)$. Apply $\ln y \le y - 1$ to $y = Q(x)/P(x)$:

$$\begin{aligned} -\sum_x P(x) \ln \frac{P(x)}{Q(x)} \;=\; \sum_x P(x) \ln \frac{Q(x)}{P(x)} \;&\le\; \sum_x P(x) \left(\frac{Q(x)}{P(x)} - 1 \right) \\ &= \sum_x Q(x) - \sum_x P(x) \;\le\; 0, \end{aligned}$$

where the last step uses $\sum_x Q(x) \le 1 = \sum_x P(x)$ (the sum over $\{x : P(x) > 0\}$ may miss some Q-mass). Negating gives the claim. Equality in $\ln y \le y - 1$ requires $y = 1$ for every x with $P(x) > 0$, i.e. $Q(x) = P(x)$ there; equality in $\sum Q \le \sum P$ then forces Q to have no mass outside $\operatorname{supp} P$. Both together give $P = Q$ on all of $\mathcal{X}$. □

Proof of Theorem 2.2. Part (a). Apply Lemma B.1 with $\mathcal{X} = \mathbb{B}^n$, $P(x_{1:n}) = \nu(x_{1:n})$, $Q(x_{1:n}) = \rho(x_{1:n})$. Both are probability distributions on $\mathbb{B}^n$ since ν and ρ are environments. The lemma gives $D_n(\nu \parallel \rho) \geq 0$ with the stated equality condition.

Part (b). For each fixed history $x_{<t}$ with $\nu(x_{<t}) > 0$, apply Lemma B.1 with $\mathcal{X} = \mathbb{B}$, $P = \nu(\cdot \mid x_{<t})$, $Q = \rho(\cdot \mid x_{<t})$ (both are probability distributions on $\mathbb{B}$):

$$\sum_{x_t \in \mathbb{B}} \nu(x_t \mid x_{<t}) \ln \frac{\nu(x_t \mid x_{<t})}{\rho(x_t \mid x_{<t})} \;\geq\; 0,$$

with equality iff $\nu(\cdot \mid x_{<t}) = \rho(\cdot \mid x_{<t})$. Multiply by $\nu(x_{<t}) \geq 0$ and sum over $x_{<t}$: this gives $d_t(\nu \parallel \rho) \geq 0$, as a sum of non-negative terms. Equality forces every $x_{<t}$ with $\nu(x_{<t}) > 0$ to contribute a vanishing inner sum, which is exactly the stated condition. □

C. Model Class $\mathcal{M}$

All results in this sheet hold for any countable class $\mathcal{M}$ and prior w_ν. The canonical **Solomonoff** choice takes $\mathcal{M}$ to be the class of all computable environments and the **Solomonoff prior**

$$w_\nu \;:=\; 2^{-K(\nu)},$$

where $K(\nu)$ is the *Kolmogorov complexity* of ν: the length of the shortest program that computes ν. The Bayesian mixture ξ under this particular prior is the *universal mixture*, written ξ_U. With this class the assumption $\mu \in \mathcal{M}$ reduces to "the universe is computable": as weak an assumption as one can make. For technical details on Kolmogorov complexity, see [HQC24, §2.7].

The class of all computable measures is countable but not enumerable (by the halting problem, you can't decide which Turing machines define total functions), so one can't even formally write $\sum_{\nu \in \mathcal{M}} \ldots$, as the decision problem of determining if $\nu \overset{?}{\in} \mathcal{M}$ is undecidable. A faithful construction of ξ_U requires expanding $\mathcal{M}$ to all *lower-semicomputable semimeasures* (Levin), at which point ξ_U becomes a semimeasure rather than a measure: $\sum_x \xi_U(x) \leq 1$ instead of $= 1$, conditionals may not sum to 1, and every other proof in this sheet gains a side-case. This also gives the nice benefit that $\xi \in \mathcal{M}$.

We dodge all of this: $\mathcal{M}$ is just *some* countable set of proper measures with $\mu \in \mathcal{M}$, and we assume $K(\nu)$ is defined for each $\nu \in \mathcal{M}$ when we need it. We never need ξ itself to be in $\mathcal{M}$, nor do we need its computability, so we don't ask. See [Hut05, §2.4.3] for the full Levin construction.

D. Best choice of $\hat{\mu} \in \mathcal{M}$.

The bound is valid for every $\hat{\mu} \in \mathcal{M}$. A tempting question: which $\hat{\mu}$ gives the tightest bound? The answer is genuinely setting-dependent.

Finite horizon n. The minimizer of the RHS is

$$\hat{\mu}_n \;:=\; \arg\min_{\nu \in \mathcal{M}} \Big\{ -\ln w_\nu \;+\; D_n(\mu \,\|\, \nu) \Big\}.$$

This is well-defined (finitely many terms, attained when $\mathcal{M}$ is finite), but the answer depends on n: for small n the prior term dominates and a high-prior coarse model wins; for large n the fit term dominates and the best per-step approximator wins. Both regimes are correct.

Asymptotic rate. A horizon-free analogue minimizes the per-step KL rate

$$\lim_{n\to\infty} \frac{1}{n} \sum_{t \le n} d_t(\mu \,\|\, \nu).$$

This is the right notion when one cares about the linear-growth slope of the bound. But it has two drawbacks: the limit need not exist for non-stationary μ (one can substitute lim inf but at the cost of clarity), and it discards the prior w_ν entirely, so the minimizer is not the same as $\hat{\mu}_n$ at any finite n, since only the slope is minimized.

Stationary μ. The per-step KLs $d_t(\mu \,\|\, \nu)$ are constant in t, so all three notions essentially agree: $\hat{\mu} = \arg\min_{\nu \in \mathcal{M}} D_{\mathrm{KL}}(\mu \,\|\, \nu)$, where the KL is taken under the (common) one-step conditional law. This is the case where "$\hat{\mu}$ = projection of μ onto $\mathcal{M}$ in KL" has unambiguous meaning.

Upshot. The bound holds for all $\hat{\mu} \in \mathcal{M}$, and one is free to take the infimum on the RHS over $\hat{\mu}$. Which $\hat{\mu}$ actually attains that infimum is a separate question whose answer depends on horizon, prior, and any structural assumptions on μ.

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